factor dependence of yields so that yield dynamics at a particular maturity are tightly bound to other maturities. The affine term structure models do this in a very tractable way so that bond yields are affine (constant plus linear) functions of risk factors. This makes the models straightforward to use in many applications from asset allocation to derivative pricing. The risk premiums in these models change over time—so the shape of the yield curve can take different shapes in recessions and expansions or in times of low or high volatility.

The first generation of term structure models employed only latent variables. That is, they described the yield curve by using only variables that were not observable macro factors and were "filtered" out from the bond yields themselves. These models did provide interpretations for what these latent factors represented by their action on the yield curve: the level, slope, and curvature factors, with the most important of these being the level factor (see Figure 9.2 and 9.3).[20] A sizeable macro-finance literature now characterizes the movements of yields and risk premiums using macro factors, including economic growth and inflation, partly inspired by Ang and Piazzesi (2003).

Monika Piazzesi and I wrote the first draft of this paper in 1998 while we were PhD students at Stanford University. At the time, we could not foresee the big bump in our citation counts that it would generate (we got lucky). We brought together the Taylor policy rule and factor dynamics for inflation and output that were widely used in macro models, with time-varying risk premium models in finance that were able to closely match bond prices. An advantage of the model is that it explicitly showed how monetary policy risk was priced in the yield curve. We have since worked on other macro-finance term structure models together, with our students, and other co-authors.

3.3. MACRO RISK PREMIUMS IN LONG-TERM BONDS

The macro-finance literature shows that macro factors play a very important part in the dynamics of the yield curve and the excess returns of bonds. Ang and Piazzesi find that the variance of bond yields at various points on the yield curve has the following attributions to output, inflation, and other (including monetary policy) factors:

	Inflation	**Output**	**Other Factors (Including Monetary Policy)**
Short End	70%	13%	17%
Middle	62%	11%	27%
Long End	32%	6%	62%

[20] Other labels are, of course, used. The canonical study of Dai and Singleton (2000), for example, labels the three factors "level," "central tendency," and "volatility." The third factor has slightly different dynamics depending on the particular affine specification employed.